

GENERATIVE AI DISCLOSURE

General Disclosure

This code was generated with the use of **ChatGPT 5.6** [Consulted on 09/26/26 and 09/27/26].

Code Attribution

The tool was fed a few prompts over the course of the assignment:

1) Generate me a C script that can parse a IPv4 out of a long string. It can additionally check for a 4 character port after a : at the end, but it is not necessary. [**Code went unused**]

2) Please generate a IPv4 Parsing script named "extractIPv4" for use in the C programming language. Requirements: Must optionally be able to catch a 4 digit port indicated by a ':' symbol. If port has 5+ digits, the entire set of parsed characters must be disregarded. Reject anything with an incorrect octet count, an empty octet, an out-of-range octet or port, a disallowed leading zero, a second colon, a colon not immediately after the fourth octet, or stray period/colons directly adjacent to an address that is otherwise valid. [**Code seen in final submission**]

I initially used the first prompt, but realized it glossed over a multitude of issues that would take some time to catch. I decided it would be quicker to prompt a new version with a higher success rate before investing time polishing and fully understanding what it does, hence the second given prompt. The initial code was replaced, though similar to what GPT's initial output was like. Many small things had to be brought in-line with what the assignment wanted. There were a few spots where values were not set to 0/-1 as instructed by the assignment guidelines. A fourth argument was generated by the initial script that needed to be removed, made so the script could support IP addresses bigger than would need to reasonably parse. The way it was generated caused it to not output a decimal value, so I wrote octaveConversion to output the expected unsigned long values the assignment looks for.

Everything before the octaveConversion() and main() functions were generated by GPT 5.6 and subsequently modified to meet criteria. The main function and small additions to extractIPv4() were student-written.

Verification Statement

I have read over the code with the intent of learning and understanding what has been produced by GAI. Multiple test cases have been used to evaluate if the resulting code responds aptly to specific scenarios.